

Economic Observability (Non-Normative)

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ABSTRACT

We introduce a non-normative formulation for observing and comparing the economic footprint of AI-mediated software construction across specification graph nodes and time windows. The goal is diagnostic: to expose inference, verification, and ownership proxies as observable quantities under a chosen pricing surface, enabling longitudinal and comparative analysis rather than optimization prescriptions.

Keywords: software engineering — AI-assisted programming — software metrics — observability — cost modeling

1. ECONOMIC OBSERVABILITY

1.1. Pricing Surface and Footprints

Let each specification node be denoted by $u \in V$, and let t denote an iteration window (e.g., a generation run, a PR cycle, or a bounded time slice).

We assume a configurable *pricing surface* for the active toolchain:

- $\mathcal{M}$ — the set of AI models available in the product.
- $\mathcal{J}$ — the set of external tools (search, build, test, OCR, etc.).
- p_m — price per token (or billable token unit) for model $m \in \mathcal{M}$.
- q_m — fixed per-call price for model m (optional; $q_m = 0$ if none).
- r_j — price per invocation for tool $j \in \mathcal{J}$ (optional).

For each node u and window t , we observe the footprint tuple

$$(T_m(u, t), N_m(u, t))_{m \in \mathcal{M}}, \quad (N_{\text{tool}, j}(u, t))_{j \in \mathcal{J}},$$

where $T_m(u, t)$ is the (billable) token volume under model m , $N_m(u, t)$ is the number of model calls, and $N_{\text{tool}, j}(u, t)$ is the number of invocations of external tool j .

1.2. Node Inference Cost

We define the *node inference cost* as the pricing-functional applied to the observed footprint:

$$C_{\text{inf}}(u, t) \equiv \sum_{m \in \mathcal{M}} (p_m T_m(u, t) + q_m N_m(u, t)) + \sum_{j \in \mathcal{J}} r_j N_{\text{tool}, j}(u, t). \quad (1)$$

This value is directly displayable on the map as currency (\$, €, etc.) under a fixed pricing surface.

1.3. Graph (Subgraph) Inference Cost

For any subgraph or selection $S \subseteq V$ (e.g., a feature branch, epic, release slice), we define:

$$C_{\text{inf}}(S, t) \equiv \sum_{u \in S} C_{\text{inf}}(u, t). \quad (2)$$

1.4. Differential Economics

Economic observability is intended to be interpreted comparatively. We define the discrete delta and drift-velocity operators for inference cost:

$$\Delta C_{\text{inf}}(u; t_k) \equiv C_{\text{inf}}(u, t_k) - C_{\text{inf}}(u, t_{k-1}), \quad (3)$$

$$DV_C(u, t) \equiv \frac{d C_{\text{inf}}(u, t)}{dt}, \quad DV_C(u, t_k) \approx \frac{C_{\text{inf}}(u, t_k) - C_{\text{inf}}(u, t_{k-1})}{t_k - t_{k-1}}. \quad (4)$$

Analogous operators apply to any node- or graph-level metric.

1.5. Verification Cost

We define a separate (configurable) verification cost functional:

$$C_{\text{verify}}(u, t) \equiv \sum_{\ell \in \mathcal{L}} s_{\ell} N_{\text{verify}, \ell}(u, t), \quad (5)$$

where $\mathcal{L}$ is the set of verifiers (CI workflows, device farm runs, property test suites, performance regressions, etc.), $N_{\text{verify}, \ell}(u, t)$ is the number of verification runs of type ℓ , and s_{ℓ} is the unit price for ℓ (or an internal cost proxy).

1.6. Total Build Cost

We define the *build cost* for a node or selection as:

$$C_{\text{build}}(u, t) \equiv C_{\text{inf}}(u, t) + C_{\text{verify}}(u, t), \quad C_{\text{build}}(S, t) \equiv \sum_{u \in S} C_{\text{build}}(u, t). \quad (6)$$

C_{build} is the direct economic footprint of producing a concrete artifact state from intent under the current toolchain configuration. It is an observed quantity under the active pricing surface, not a structural prediction about future maintenance or change cost.

1.7. Change-Cost Proxy (Graph-Based)

We introduce a non-normative *change-cost proxy* to quantify how costly future modifications may become due to graph coupling and instability. Let $\text{WCR}_k(u)$ denote the weighted change radius within hop distance k , and let $\text{SVC}(u)$ and $\text{CSI}(u)$ denote intent verifiability and checkpoint stability, respectively (defined elsewhere).

$$C_{\text{change}}(u) \equiv \kappa \text{WCR}_k(u) \frac{1}{\text{SVC}(u) + \varepsilon} \frac{1}{\text{CSI}(u) + \varepsilon}, \quad (7)$$

where κ is a project-calibration coefficient and $\varepsilon > 0$ stabilizes the expression. This quantity is diagnostic: its meaning lies in relative comparisons and longitudinal drift.

Unlike C_{inf} , C_{verify} , and C_{build} , this quantity is not observed spend. It is a structural proxy derived from graph topology, verifiability, and stability signals.

1.8. Total Ownership Proxy

Finally, we define a composite *ownership proxy* over a selection S :

$$C_{\text{own}}(S) \equiv C_{\text{build}}(S) + \lambda \sum_{u \in S} C_{\text{change}}(u), \quad (8)$$

where λ encodes the chosen support horizon (how much weight is assigned to the future change-cost proxy). C_{own} is not an estimate guarantee and should not be interpreted as direct monetary spend. It is an economic lens for comparing structural alternatives and observing drift under AI-mediated construction.

1.9. Interpretation Note

All economic quantities are non-normative observables. Absolute values depend on pricing, tooling, and policy configuration. The recommended mode of use is comparative: deltas (Δ), drift velocities (DV), distribution shifts across the graph, and cost bands under controlled configurations.

Every reported economic artifact should preserve its provenance: pricing surface identifier, model and tool versions, time window, currency or unit convention, and whether each displayed quantity is observed spend or a derived proxy.

1.10. Map View Encoding (UI Mapping)

We treat the map as a *projection* of observability signals onto visual channels. Let $x(u) \in \mathbb{R}^2$ denote the fixed layout coordinate of node u in the map view. Under a selected scope $S \subseteq V$ and time window t , the UI encodes:

- **Fill color (economic heat):** node fill is mapped to instantaneous inference cost

$$\text{Color}(u) \Leftarrow C_{\text{inf}}(u, t), \quad (9)$$

typically via a monotone colormap with optional banding.

- **Node radius (growth / pressure):** node size reflects a chosen differential economic signal, by default

$$\text{Radius}(u) \Leftarrow |\Delta C_{\text{inf}}(u; t_k)| \quad \text{or} \quad \text{Radius}(u) \Leftarrow |\text{DV}_C(u, t)|. \quad (10)$$

- **Halo / influence ring (blast radius):** an outer ring visualizes change impact at hop distance k ,

$$\text{Halo}(u) \Leftarrow \text{WCR}_k(u), \quad (11)$$

where the ring radius or intensity increases with weighted reachability.

- **Border style (confidence / stability):** border opacity or dash pattern encodes verification and stability,

$$\text{Border}(u) \Leftarrow (\text{SVC}(u), \text{CSI}(u, t)), \quad (12)$$

where low SVC (weak verifiability) and low CSI (unstable checkpoints) are rendered with reduced opacity and/or higher visual “noise”.

- **Edge emphasis (coupling cost flow):** when edges are shown, thickness or glow corresponds to the marginal cost transfer associated with coupling, e.g.

$$\text{EdgeWeight}(u, v) \Leftarrow \Delta C_{\text{inf}}(u; t_k) \mathbf{1}_{(u \rightarrow v)} \quad \text{or} \quad w(u, v), \quad (13)$$

where $w(u, v)$ is an application-defined coupling weight (dependency strength, co-change frequency, or interface-touch severity).

- **Scope economics (non-normative totals):** the current selection S reports aggregated economics

$$\text{ScopeTotals}(S) \Leftarrow (C_{\text{inf}}(S, t), C_{\text{build}}(S, t), C_{\text{own}}(S)), \quad (14)$$

always accompanied by the active pricing surface and time window.

The mapping is explicitly *lens-based*: the user may choose which economic signal drives each channel (color, radius, halo, border), while the system preserves comparability by fixing the pricing surface and time window for a given analysis session.